PROJECT RISK MANAGEMENT

WORK INSTRUCTION

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O1	11-Oct-19	Wisanu P.	Chotechuang K.	Worasit L.	Sangiam C.
Rev	Date	Prepared By	Reviewed By	Approved By	Quality Review By

Contents

1 OBJECTIVE

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The purpose of this Work Instruction (WI) is to define the method by which project execution risks, arising from the proposal phase through the execution phase, are managed in a timely manner to ensure the achievement of project objectives related to cost, schedule, quality, safety, and contractual performance.

2 ABBREVIATIONS AND DEFINITIONS

Project Risk	An uncertain event or condition that, if it occurs, has a negative effect on one or more project objectives.
Risk Register	A repository in which outputs of risk management processes are recorded. Information in the risk register can include the person responsible for managing the risk, probability, impact, risk score, planned risk responses, and other information used to get a high-level understanding of risks.

3 SCOPE

This WI applies from contract award through project close-out for all EPC projects.

4 RESPONSIBILITIES

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Project Manager (PM) shall be accountable for establishing, implementing, monitoring, and escalating project risks to protect project objectives for safety, cost, schedule, quality, and contractual compliance. The key responsibilities include:

- Enforce the project risk management framework in line with corporate policy and contract requirements.
- Ensure risks are identified across technical, commercial, HSE, schedule, cost, interface, and execution
- Confirm that contractual and commercial risks (LDs, performance guarantees, changing mechanisms) are fully captured.
- Validate risk severity, likelihood, and exposure, including cost and schedule impacts
- Make informed decisions in a timely manner while balancing risk, cost, schedule, quality, and safety
- Approve risk response strategies (avoid, reduce, transfer, accept, exploit)

- Ensure mitigation actions are:
  - Clearly defined
  - Assigned to accountable owners
  - Resourced and time-bound
- Monitor Discipline Leads' execution of mitigation actions
- Escalate unresolved or deteriorating risks to Project Director / Steering Committee
- Chair or sponsor regular risk review meetings
- Ensure the project Risk Register is:
  - Up to date
  - Actively used as a management tool
  - Linked to schedule, cost forecast, and change control
- Approve risk-based trade-offs (e.g. acceleration vs. increased cost risk)
- Encouraging senior management support for project risk management activities. Escalating identified risks to senior management where appropriate such risks include any which are outside the authority or control of the project manager, any which require input or action from outside the project
- Determining the acceptable levels of risk for the project in consultation with stakeholders
- Promoting the project risk management process
- Overseeing risk management by subcontractors and suppliers
- Regularly updating and reporting risk status to key stakeholders with recommendations for strategic decision making and actions to maintain acceptable risk exposure
- Monitoring the efficiency and effectiveness of the project risk management process
- Documenting lesson learned relate to project risk management

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Project Engineer (PE) shall be responsible for facilitating the project risk management, e.g. prepare project Risk Register, organize the risk analysis meetings and update the project Risk Register monthly.

Discipline Leads shall be accountable for the timely identification of risks and the implementation of the approved risk mitigation actions within their areas.

The responsibility flowchart is provided below:

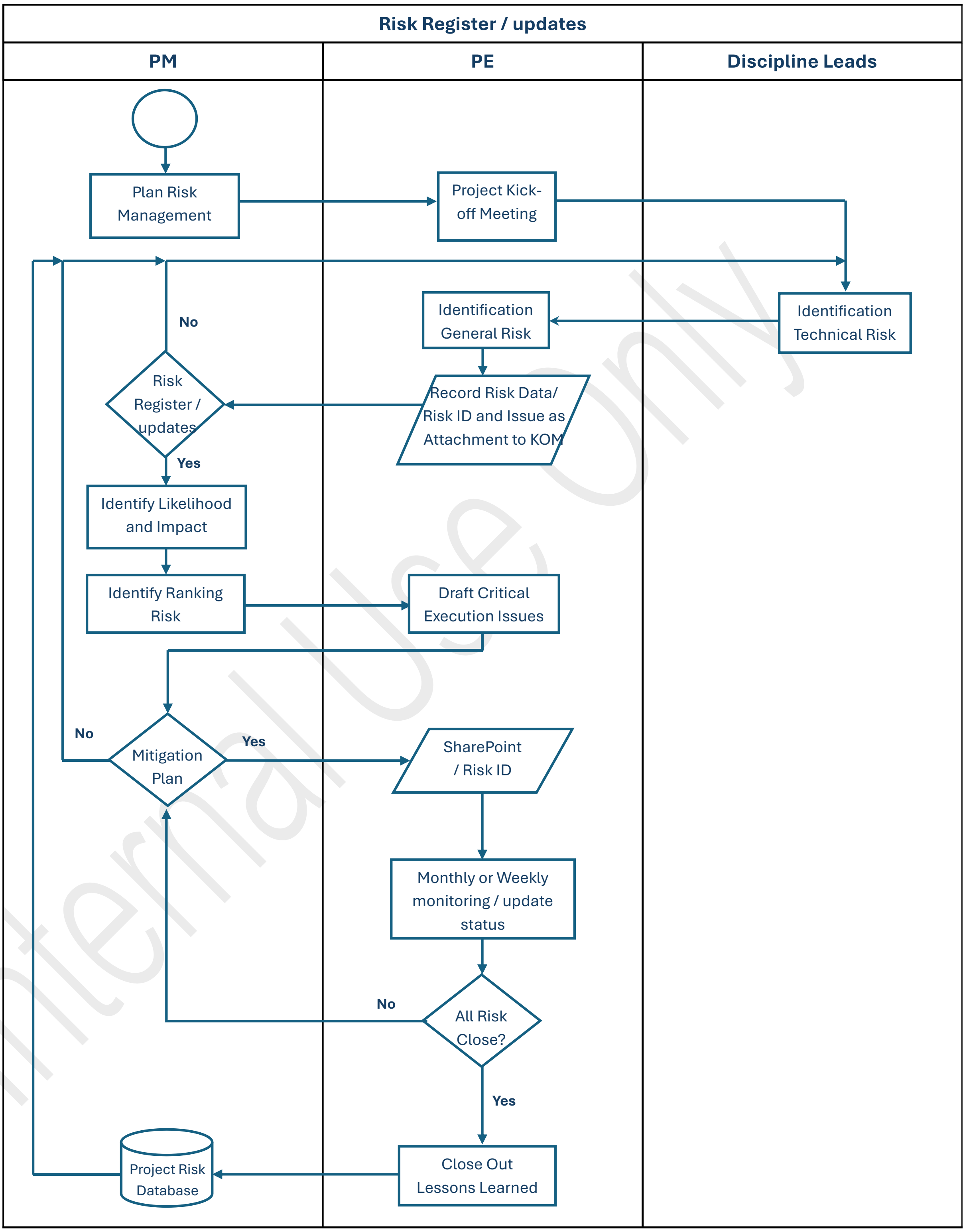

Fig. 4-1 Responsibility Flowchart

5 PROJECT RISK MANAGEMENT PROCESSES

Project risk management provides an approach by which uncertainty can be understood, assessed and managed within project. The Project Risk Management process comprises of six processes as follows, and the project risk management process diagram is shown in Fig. 5-1.

5.1 Plan Risk Management

The process of defining how to conduct risk management activities for a project. The key benefit of this process is that it ensures that the degree, type, and visibility of risk management are proportionate to both risks and the importance of the project to the organization and other stakeholders. The Plan Risk Management process should begin when a project is conceived and should be completed early in the project.

5.2 Identify Risks

The process of identifying project risks as well as sources of overall project risk and documenting their characteristics.

5.3 Perform Qualitative Risk Analysis

The process of prioritizing project risks for further analysis or action by assessing their probability of occurrence and impact as well as other characteristics.

5.4 Perform Quantitative Risk Analysis

The process of numerically analyzing the combined effect of identified project risks and other sources of uncertainty on overall project objectives.

5.5 Plan Risk Responses

The process of developing options, selecting strategies, and agreeing on actions to address overall project risk exposure, as well as to treat project risks.

There are four risk response strategies:

1) **Avoid a Threat or Exploit an Opportunity**

   It involves taking actions to ensure either that the threat cannot occur or can have no effect on the project, or that the opportunity will occur and the project will be able to take advantage of it.

2) **Transfer a Threat or Share an Opportunity**

   It involves transference of a threat or sharing the opportunity to the third party

3) **Mitigate a Threat or Enhance an Opportunity**

   This strategy is the most widely applicable and widely used. It involves identifying the actions that decrease the probability and/or the impact of a threat and increase the probability and/or the impact of an opportunity.

4) **Accept a Threat or an Opportunity**

   This strategy applies when the other strategy are not considered applicable or feasible. Taking no actions unless the threat occurs, in which case contingency or fallback plan may be developed ahead of time, to be implemented if the threat is present.

5.6 Monitor and Control Risks

The process of monitoring the implementation of agreed-upon risk response plans, tracking identified risks, identifying and analyzing new risks, and evaluating risk process effectiveness throughout the project.

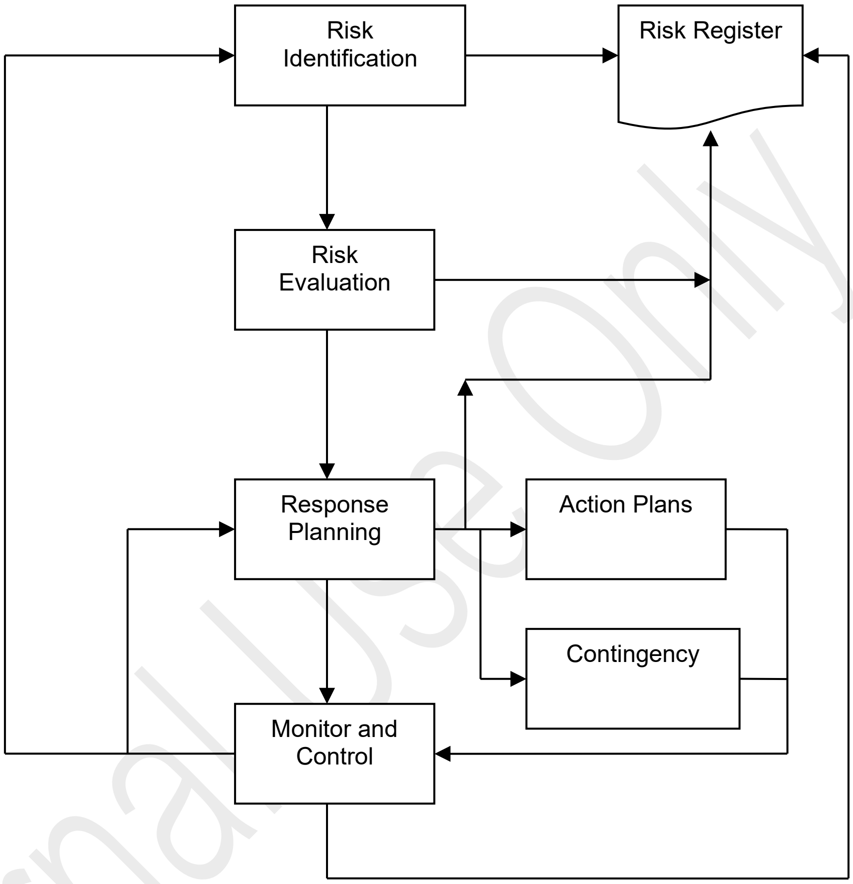

Fig. 5-1 Project Risk Management Processes

6 CRITICAL SUCCESS FACTORS

The general criteria for success include:

- Recognize the value of risk management
- Individual commitment/Responsibility - risk management is everybody's responsibility
- Open and honest communication - everyone should be involved in the risk management process.
- Organizational commitment - risk management may require a higher level of managerial support
- Risk effort scaled to Project – project risk management activities should be consistent with the value of the project, its level of project risk, its scale and other constraints
- Integration with project management – successful project risk management requires the correct execution of the other project management processes

7 INSTRUCTION

7.1 Proposal Stage

The nominated PM shall be responsible for leading the team to:

1) **Identify Project Risks**

Nominated Project Engineer shall be responsible for facilitating the registration of all project risks into project Risk Register in SharePoint at: *https://pttgcgroup.sharepoint.com/sites/EPCAProjectEngineer/Lists/DataRisk/*

Each identified project risk should relate to at least one project objective, e.g. time cost, quality, scope, etc. Some risk may affect more than one objective, and it should be described at the level of detail at which it can be assigned to a single risk owner. Describing each risk using three-part statement in the form: "As a result of…(cause), risk may occur, which would lead to…(effect)"

The risk identification should be completed prior to proposal review with management.

2) Perform Qualitative Risk Analysis

The consequence severity definitions provided in Attachment EPM-03-014 AT1 shall be reviewed and updated as necessary to reflect the specific nature and context of the project under assessment. Similarly, the likelihood definitions provided in Attachment EPM-03-014 AT2 shall be reviewed and adjusted, where required, to ensure they align with the project's characteristics.

Using the likelihood and consequence severity criteria, the project risks shall then be qualitatively evaluated and prioritized into the following categories for selecting the effective and appropriate risk response strategy:

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- Extreme ➔ Serious risk and need action & risk reduction plan immediately
- Critical ➔ Immediate attention / action required to mitigate the risk.
- Significant ➔ Action should be taken to compensate for the risk.
- Low ➔ Action as soon as practicable and should be taken to monitor the risk.
- Very Low ➔ Low priority, possibly no action required or routine acceptance of the risk.

Based on the likelihood of occurrence and the severity of consequences on the project objectives, the Project Risk Matrix will be automatedly developed.

Following the implementation of mitigation measures, the likelihood and consequences of each identified risk shall be reviewed and revised as necessary. All residual risks shall then be reassessed, and the results recorded in project Risk Register.

3) Plan Risk Responses

Effective risk response actions shall be defined for each identified risk, including assignment of a Risk Action Owner, establishment of appropriate response deadlines, and specification of current risk status. The Risk Action Owner is responsible for ensuring that the agreed risk responses are implemented as planned and in a timely manner.

Risk responses, once implemented, may generate additional risks or leave residual risks. Any additional or residual risks shall be identified, assessed, and managed using the same process applied to the initially identified risks.

In general, risk responses should be appropriate, timely, cost-effective, feasible, achievable, clearly assigned, formally agreed, and accepted by relevant stakeholders.

All risk response actions, ownership, deadlines, status, and residual risks shall be documented and maintained in the project Risk Register.

7.2 Project Execution Stage

PM shall be responsible for leading the project team to:

1) Update Project Risk Register

Discipline Leads shall be accountable for updating or identifying new risks within their areas.

Project Engineer shall be responsible for facilitating the update of, and addition of new risks to project Risk Register established in SharePoint during the proposal stage. This includes assessing newly identified risks and residual risks using qualitative risk analysis and defining effective response actions, following the same methodology applied during proposal preparation, as detailed in Section 7.1 above.

PM shall ensure that their Risk Register is established, updated monthly and maintained in SharePoint as the central project risk database

PM shall communicate the updated project Risk Register in the internal kick-off meeting or weekly meeting and issue as an attachment to the Note of Meeting.

2) Monitor and Control Risks

The primary objectives of risk monitoring and controlling are to track identified risks, monitor residual risks, identify new risks, ensure that risk responses are executed at the appropriate time, and evaluate their effectiveness throughout the project life circle.

As the project progresses, PM shall ensure that risk reassessment, including risk identification, qualitative analysis and risk responses is repeated at the reasonable intervals or in response to the project events without generating excessive admin. overhead.

At the end of the project, PM shall ensure that the status of every risk in the project Risk Register is closed and the actual data is recorded for future use.

8 REFERENCE DOCUMENTS

- Practice Standard for Project Risk Management published by Project Management Institute, Inc. 2019

9 ATTACHMENTS

Attachment EPM-03-014 AT1 Consequence Severity Definitions

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Attachment EPM-03-014 AT2 Probability Definitions

ATTACHMENT EPM-03-014AT1 – TYPICAL RISK REGISTER

GCME

Typical Project Risk Register

As of............

Project:				
Document no.:				
Rev.	Date	Prepared	Checked	Approved
1				

Identify Risks		Assess Risks			Treat Risks/ Risk Response	Control risks					Stakeholder	Time Frame			Status	Comment
Risk ID	Risk Description	Possible Effect Cost/ Time/ Quality/ HSE (C/T/Q/H)	Probability Insignificant/ Minor/Moderate/Major/Severe (1/2/3/4/5)	Serverity Of Impact Rarely/Unlikely /Occasional/Likely/Most Likely (1/2/3/4/5)	Mitigation Strategy A/T/M/AC	Action to Control & Mitigate	Probability Insignificant /Minor/Moderate/Major/Severe (1/2/3/4/5)	Serverity Of Impact Rarely/Unlikely/Occasional /Likely/Most Likely (1/2/3/4/5)	Cost to Mitigate	Probability of Success	Owner	Raised Date	Deadline Date	Finished Date	Status	Comment
T1																
No.	Risk Description	C/T/Q/H	Probability	Serverity Of Impact	Mitigation Strategy	Action to Mitigate	Probability	Serverity Of Impact	CTM	POS	Owner				Status	Status
1	Process design															
							2									
2	Civil & Structure design															
3	Piping design															
4	Mechanical design															
5	Electrical design															
6	Instrument design															
7	Procurement															
8	Construction															
9	Safety,Security Healthand Environmental															
10	Other															

Legend:

Probability - The likelihood that a risk will occur,

Probability Description	Score
Almost certain to occur	5
Probably will occur	4
50/50 chance	3
Probably not	2
Very Unlikely	1

Risk Ranking Matrix

Probability of occurrence							
5	Severe	Low	Significant	Critical	Extreme	Extreme
4	Major	Low	Significant	Critical	Critical	Extreme
3	Moderate	Low	Low	Significant	Critical	Critical
2	Minor	Very Low	Low	Low	Significant	Significant
1	Insignificant	Very Low	Very Low	Low	Low	Low
Level		Rarely	Unlikely	Occasional	Likely	Most Likely
		1	2	3	4	5
		Serverity Of Impact				

=>

The evaluation of the severity and probability

Probability of occurrence							
5	Very high				Must mitigate		
4	High						
3	Medium			Should mitigate			
2	Low	Consider accepting					
1	Very Low						
Level		Very Low	Low	Medium	High	Very high	Very high
		1	2	3	4	5	5
		Serverity Of Impact					

Serverity Of Impact - The effect that risk will have on the project if it occurs,

Impact or severity	Score		SAFETY	EQ COST (Baht)
Very severe, few alternatives or work arounds	5	>>==>>>	Fatality	>50M
Pretty severe, complex alternatives/ work arounds	4	>>==>>>	LTI	<50M
Many reasonable alternatives	3	>>==>>>	Multiple Medical Treatments	<5M
Alternatives within plan, cost	2	>>==>>>	First Aid	<500K
Limited, no alternatives needed	1	>>==>>>	Minor incidents without injury	<100K

Strategy- The strategy to minimise the risk.

Strategy Description	Abbreviation
Avoid: Risk avoidance is to eliminate the risk or protect the project objectives from its impact	A
Transfer: Risk transfer is seeking to shift the risk to a third party	T
Mitigate: Mitigation is reducing the probability and/or severity of an adverse risk event to and acceptable level	M
Accept: When the impact of a potential risk event is relatively small, the project team may decide to simply accept the risk	AC

Cost to Mitigate (CTM) - The Cost to mitigate the risk.

High	H
Medium	M
Low	L

Probability of Success (POS) - The Probability of Success

High	H
Medium	M
Low	L

ATTACHMENT EPM-03-014AT2 – Probability Definitions

	Financial	Safety, Health & Environment		Supplier/Customers/ Partners		Law and Regulation	Reputation/Image		Deviation
	Schedule Impact Exposure to loss from a project not meeting its schedule objectives. **Cost** Probability of loss due to cost overrun (Also includes decreases in revenue/EBITDA.)	**Safety and Health** A situation or circumstance that includes the possibility of injury to someone or of damage to property.	**Environment** Actual or potential threat of adverse effects on living organisms and environment by effluents, emissions, wastes, resource depletion, etc., arising out of an organization's activities.	**Supplier**	**Customers**		**Reputation & Image** The risk that a company will lose potential business because its character has been called into question.	**Negative News**	
5 **Severe**	**Project Delay** : **X > 20%** of project timeframe or <u>> 6 months</u> **Cost Overrun** : **X > 20%** of total project cost **EBITDA** : Having **> 20%** impact on 1st full-year EBITDA of the project	**Single or Multiple Fatalities** - permanent injury or disability, many disabilities or death > 1 person. Severely affects health. The communities surrounding the factory were affected by occupation and had to evacuate the area for > 1 year. There are employees or contractors to strike, stop work.	**Significant** harm with widespread effect. Recovery measurable longer than 1 year to recover. Limited prospect of full recovery.	Partner : Cancellation of the contract for the sale of goods and services Affects sales > 50% of sales/year	Customer: Towards the (Boycott) purchase of goods and services Affects sales > 50% of sales/year	[Black Swan] The company was delisted from the stock exchange. Illegal criminal or civil law and order the company to stop doing business. License revoked or temporarily suspend.	[Black Swan] reputation and image of the company severe damage severely affected confidence And it is a national issue that the relevant ministries or government agencies are monitoring. Review and jointly take corrective action. Disaster with potential to lead to collapse of the project. Long term regional impact.	Received complaints as negative news There is news in foreign media.	Below target >50%
4 **Major**	**Project Delay: X ≥ 10%** of project timeframe or <u>**> 3 months**</u> **Cost Overrun: X ≥ 10%** of total project cost •EBITDA : Having **X ≥ 10%** impact on 1st full-year EBITDA of the project	**Major or Multiple Injuries** - permanent injury or disability or to the point of disability or death. Affects a lot of health. There are protests or formal complaints.	**Significant** harm with local effect. Causes a lot of damage to the environment. It takes more than 6-12 months to recover.	Losing major partners Major partners are unable to deliver goods or services.	Lose a big customer Trading volume of major products decreased by more than 20% Major customers are unable to pay for products.	Illegal, criminal or civil law. or in violation of the Company's regulations, criminal penalties, imprisonment and fines.	Reputation and image damage was so severe that it resulted in the lack of confidence in outsiders or organizations. A critical event which requires extraordinary management effort. Localised, long term impact with unmanageable outcomes.	Received complaints as negative news. There is news in the media at the national level.	Below target >30%-50%
3 **Moderate**	**Project Delay: 6% ≤ X < 10%** of project timeframe or <u>**< 3 months**</u> **Cost Overrun: 6% ≤ X < 10%** of total project cost **EBITDA:** Having **6% ≤ X < 10%** impact on 1st full-year EBITDA of the project	**Serious Injuries** require Medical Treatment, or Lost Work Case. Moderate impact on health and long-term morale	**Moderate** harm with possible wider effect. Recovery measurable 3-6 months to recover.	Major partners Make a letter expressing dissatisfaction (Official Complain) Major partners are able to deliver some products or services.	Major customers Make a letter expressing dissatisfaction (Official Complain) Trading volume of major products decreased by 10-20% Major customers can deliver some products or services.	May make it unlawful. or the regulations of the company (Non-compliance with laws/Articles of association).	Affects the reputation and image of the Company Moderately causes dissatisfaction from the community or the public. A serious event which requires additional management effort. Localised, long term impact but manageable.	It's negative news. There is news released in the media at the provincial and regional level.	Below target >10%-30%
2 **Minor**	**Project Delay: 3% ≤ X < 6%** of project timeframe or <u>**< 2 months**</u> **Cost Overrun: 3% ≤ X < 6%** of total project cost as BOD approval **EBITDA:** Having **3% ≤ X < 6%** impact on 1st full-year EBITDA of the project	**Minor Injuries** require Medical Treatment with/or Restricted Work Case. Low impact on health and short-term morale	Less damage to the environment. Recovery measurable less than 3 months to recover.	Verbal complains. Retail partners are unable to deliver goods or services.	Customers express dissatisfaction (Verbal complains). Trading volume of major products decreased by 5-10%. Retail customers are unable to pay for products. Retail partners are unable to deliver goods or services.	May cause non-conformance. or standards within the company (Non-conformance).	Affects the reputation and image of the company slightly, can be corrected in a short period of time. An adverse event which can be absorbed with some management effort. Localised, short term impact.	Receive complaints from the community through informal company channels It's negative news. There is news in the local media.	Below target >5-10%
1 **Insignificant**	**Project Delay: X < 3%** of project timeframe or <u>**< 1 month**</u> **Cost Overrun: X < 3%** of total project cost **EBITDA:** Having **X < 3%** impact on 1st full-year EBITDA of the project	**First Aid Case** There were no injuries or minor injuries at first aid level. No effect or minimal impact on health and morale.	**No impact** on baseline environment. Localized to point source. No recovery required.	Retail partners Dissatisfied, complained back. Retail partners can deliver some products or services.	Retail customers Dissatisfied, complained back. Product trading volume dropped below 5% Retail customers can pay for some products.	Failure to comply with laws or regulations. but in a way that can be easily fixed.	Little impact on reputation and image of the company. Impact can be absorbed through normal activity. Localised temporary impact.	It's not a trend in the local media.	Below target <5%

		LIKELIHOOD (FREQUENCY)				
		1	**2**	**3**	**4**	**5**
		Rarely	**Unlikely**	**Occasional**	**Likely**	**Most Likely**
Probability	**Probability of Fture Events**	< 5%	5% - 10%	>10% - 25%	>25% - 50%	> 50%
		Highly unlikely to occur on this project	Given current practices and procedures, this event is unlikely to occur on this project	This event has occurred on a similar project	Event is Likely to occur on this project	Event is very likely to occur on this project, possibly several times
	Probability of Past Events	Never happened or may occur over a period of more than 3 years	Has happened once in the past 3 years in an industry or business or system similar to that of GCME.	Has happened more than once in the past 3 years in an industry or business or system similar to that of GCME.	Has happened 1 time locally or more times in the GCME in the past 1 year.	Has happened more than 1 time locally in the past 1 year in GCME
Experience / Capability to	**Work process/ Ability to Control**	The workflow is simple. and less chance of error There is a good audit or control. Has a very good control ability.	The workflow is more complicated. But there is also good monitoring or control. Have good control ability	The work process is complicated. But there is a check or moderately controlled. Have some or moderate control ability	The work process is complicated. and there are few checks or still have little control. Little control ability	The workflow is very complicated. and no inspection or has never been regulated. There is little or no control ability.
	Past Experience and Success	Have direct experience in that business or job	Very experienced in a similar business or type of work	Have some or moderate experience in that business or job	little experience in that business or job	Little or no experience in that business or job

10 HISTORY RECORD

Rev.	Page	Description Edit	Date Issue
O1	All	Issued for Review / Approval	11-Oct-19
F1	All	Issued for Final	30-Oct-19
F2	P.5, P.7	Project's risk assessment process is worth less than 200 MB	3 Sep 2021
F3	All	Issued for Review / Approval	15-Dec-25